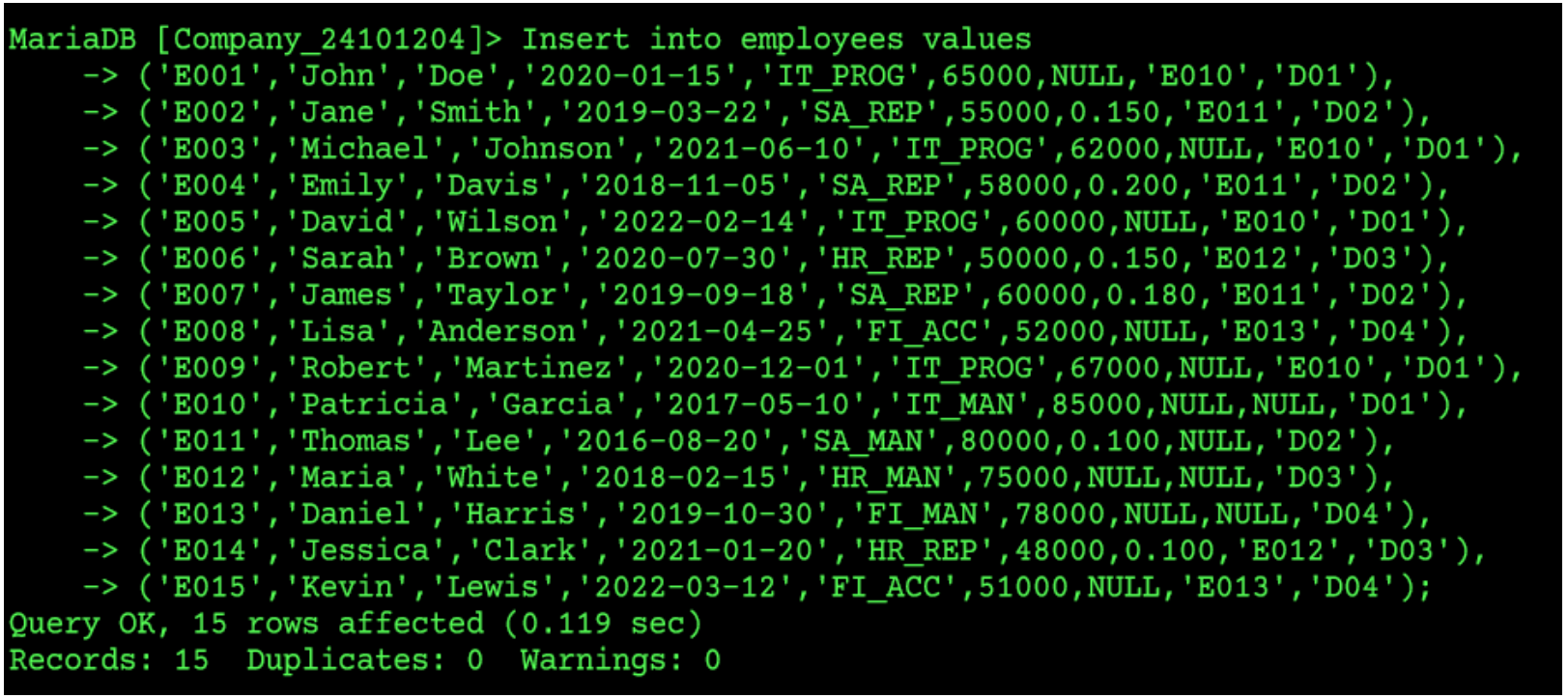
CSE370: Database Systems

Assignment 02 | Spring 2026

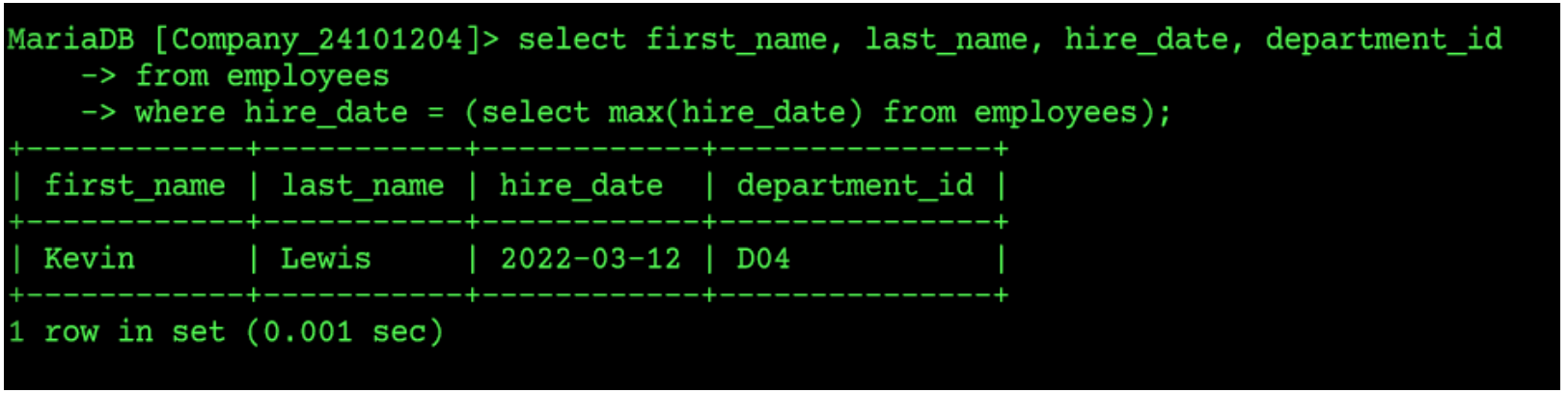
No 1 Query (as Plain Text)	`CREATE TABLE employees`
	`(`
	`  e_id CHAR(4),`
	`  first_name VARCHAR(20),`
	`  last_name VARCHAR(20),`
	`  hire_date DATE,`
	`  job_id CHAR(10),`
	`  salary INT,`
	`  commission_pct DECIMAL(5,3),`
	`  manager_id CHAR(4),`
	`  department_id CHAR(3)`
	`);`

No 1 SS (of Query & Output in Shell)	`Database changed`
	`MariaDB [Company_24101204]> CREATE TABLE employees`
	`    -> (`
	`    ->   e_id CHAR(4),`
	`    ->   first_name VARCHAR(20),`
	`    ->   last_name VARCHAR(20),`
	`    ->   hire_date DATE,`
	`    ->   job_id CHAR(10),`
	`    ->   salary INT,`
	`    ->   commission_pct DECIMAL(5,3),`
	`    ->   manager_id CHAR(4),`
	`    ->   department_id CHAR(3)`
	`    -> );`
	`Query OK, 0 rows affected (0.185 sec)`

No 2 Query (as Plain Text)	Insert into employees values ('E001','John','Doe','2020-01-15','IT_PROG',65000,NULL,'E010','D01'), ('E002','Jane','Smith','2019-03-22','SA_REP',55000,0.150,'E011','D02'), ('E003','Michael','Johnson','2021-06-10','IT_PROG',62000,NULL,'E010','D01'), ('E004','Emily','Davis','2018-11-05','SA_REP',58000,0.200,'E011','D02'), ('E005','David','Wilson','2022-02-14','IT_PROG',60000,NULL,'E010','D01'), ('E006','Sarah','Brown','2020-07-30','HR_REP',50000,0.150,'E012','D03'), ('E007','James','Taylor','2019-09-18','SA_REP',60000,0.180,'E011','D02'), ('E008','Lisa','Anderson','2021-04-25','FI_ACC',52000,NULL,'E013','D04'), ('E009','Robert','Martinez','2020-12-01','IT_PROG',67000,NULL,'E010','D01'), ('E010','Patricia','Garcia','2017-05-10','IT_MAN',85000,NULL,NULL,'D01'), ('E011','Thomas','Lee','2016-08-20','SA_MAN',80000,0.100,NULL,'D02'), ('E012','Maria','White','2018-02-15','HR_MAN',75000,NULL,NULL,'D03'), ('E013','Daniel','Harris','2019-10-30','FI_MAN',78000,NULL,NULL,'D04'), ('E014','Jessica','Clark','2021-01-20','HR_REP',48000,0.100,'E012','D03'), ('E015','Kevin','Lewis','2022-03-12','FI_ACC',51000,NULL,'E013','D04');

No 2 SS (of Query & Output in Shell)	

No 3 Query (as Plain Text)	`select first_name, last_name, hire_date, department_id`
	`from employees`
	`where hire_date = (select max(hire_date) from employees);`

No 3 SS (of Query & Output in Shell)	

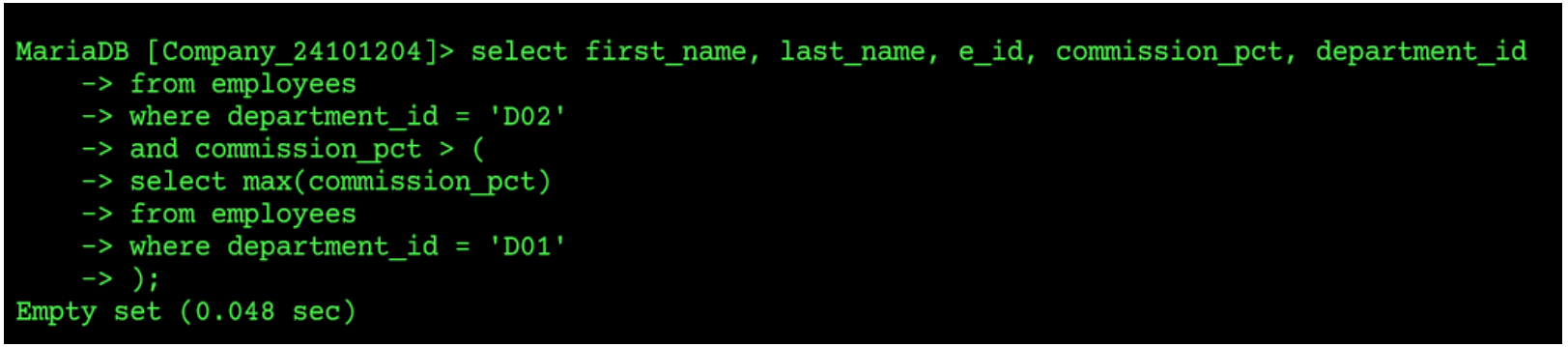
No 4 Query (as Plain Text)	`select first_name, last_name, e_id, salary, department_id`
	`from employees e`
	`where salary = (`
	`select max(salary)`
	`from employees`
	`where department_id = e.department_id`
	`);`

No 4 SS
(of Query & Output in Shell)

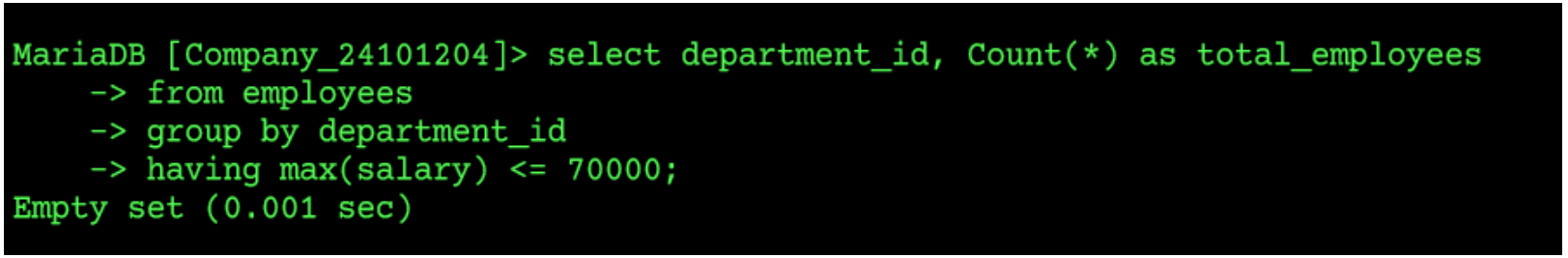
```
MariaDB [Company_24101204]> select first_name, last_name, e_id, salary, department_id
    -> from employees e
    -> where salary = (
    -> select max(salary)
    -> from employees
    -> where department_id = e.department_id
    -> );
+------------+-----------+------+--------+---------------+
| first_name | last_name | e_id | salary | department_id |
+------------+-----------+------+--------+---------------+
| Patricia   | Garcia    | E010 |  85000 | D01           |
| Thomas     | Lee       | E011 |  80000 | D02           |
| Maria      | White     | E012 |  75000 | D03           |
| Daniel     | Harris    | E013 |  78000 | D04           |
+------------+-----------+------+--------+---------------+
4 rows in set (0.001 sec)
```

No 5 Query
(as Plain Text)

```
select first_name, last_name, e_id, commission_pct, department_id
from employees
where department_id = 'D02'
and commission_pct > (
select max(commission_pct)
from employees
where department_id = 'D01'
);
```

No 5 SS (of Query & Output in Shell)	

No 6 Query (as Plain Text)	`select department_id, Count(*) as total_employees`
	`from employees`
	`group by department_id`
	`having max(salary) <= 70000;`

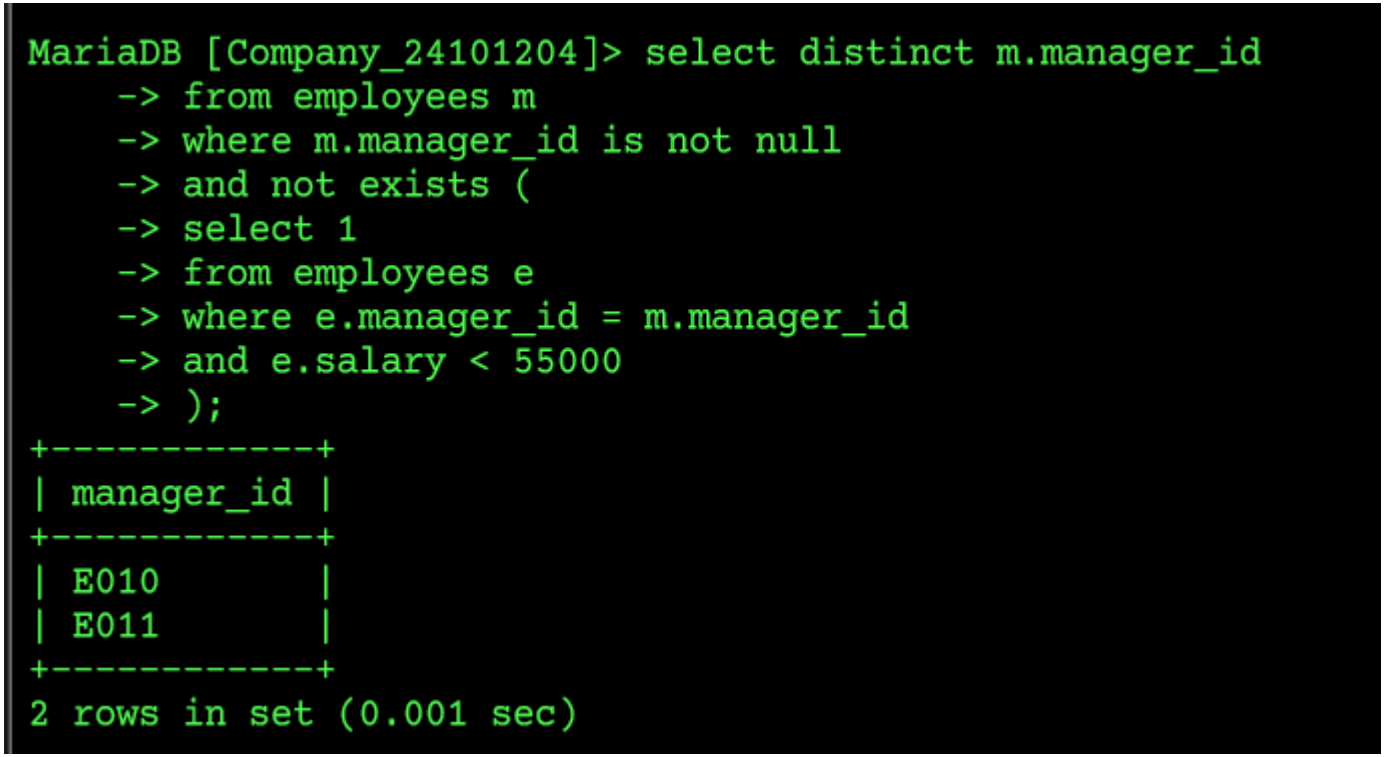
No 6 SS (of Query & Output in Shell)	

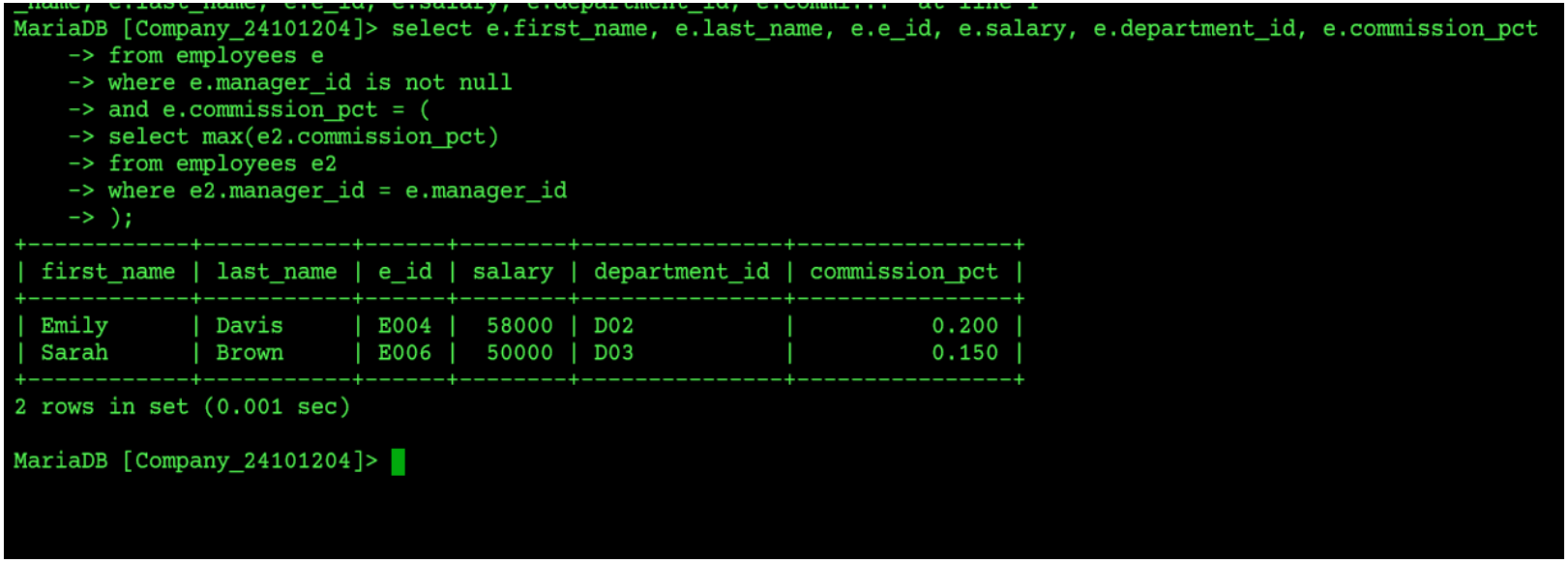
No 7 Query (as Plain Text)	`select distinct e1.department_id, e1.job_id, e1.salary`
	`from employees e1`
	`where exists (`
	`select 1 from employees e2`
	`where e2.department_id = e1.department_id`
	`and e2.job_id <> e1.job_id`

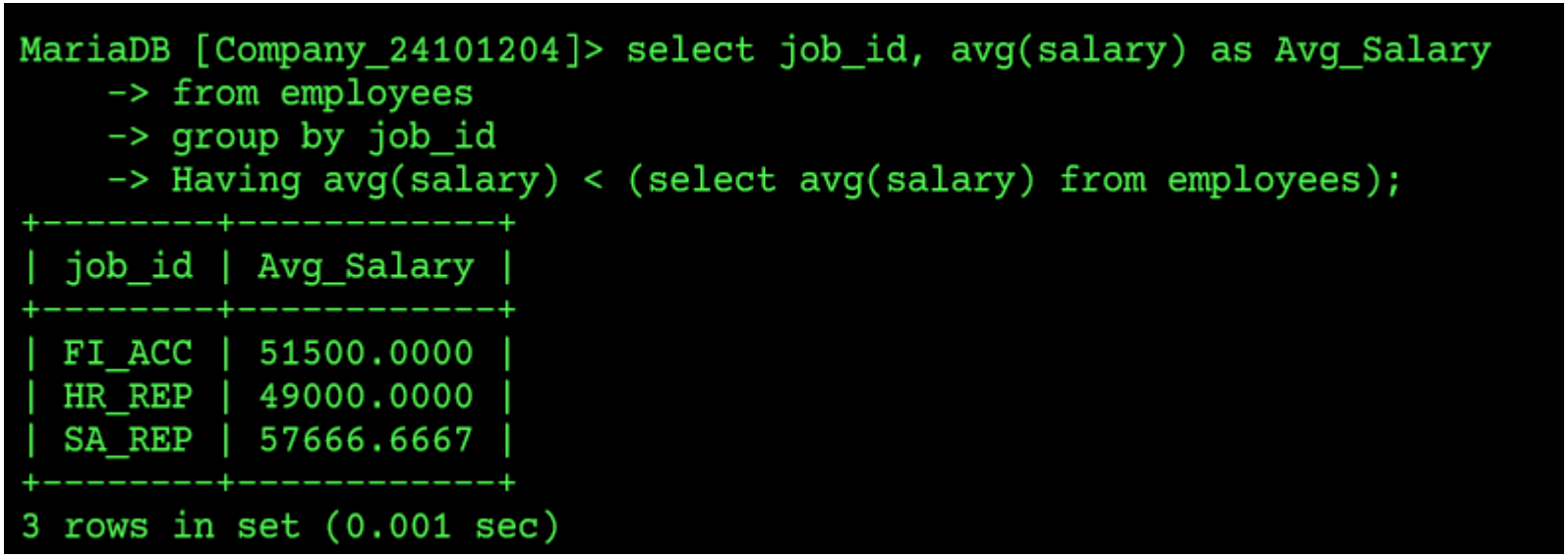
	`and e1.salary < e2.salary` `);`

No 7 SS
(of Query & Output in Shell)
	`MariaDB [Company_24101204]> select distinct e1.department_id, e1.job_id, e1.salary`
	`-> from employees e1`
	`-> where exists (`
	`-> select 1 from employees e2`
	`-> where e2.department_id = e1.department_id`
	`-> and e2.job_id <> e1.job_id`
	`-> and e1.salary < e2.salary`
	`-> );`
	`+---------------+---------+--------+`
	`| department_id | job_id  | salary |`
	`+---------------+---------+--------+`
	`| D01 | IT_PROG | 65000 |`
	`| D01 | IT_PROG | 62000 |`
	`| D01 | IT_PROG | 60000 |`
	`| D01 | IT_PROG | 67000 |`
	`| D02 | SA_REP | 55000 |`
	`| D02 | SA_REP | 58000 |`
	`| D02 | SA_REP | 60000 |`
	`| D03 | HR_REP | 50000 |`
	`| D03 | HR_REP | 48000 |`
	`| D04 | FI_ACC | 52000 |`
	`| D04 | FI_ACC | 51000 |`
	`+---------------+---------+--------+`
	`11 rows in set (0.001 sec)`

No 8 Query **(as Plain Text)**	`select distinct m.manager_id` `from employees m`

	where m.manager_id is not null and not exists (select 1 from employees e where e.manager_id = m.manager_id and e.salary < 55000);
No 8 SS (of Query & Output in Shell)	
No 9 Query (as Plain Text)	select e.first_name, e.last_name, e.e_id, e.salary, e.department_id, e.commission_pct from employees e

	where e.manager_id is not null and e.commission_pct = (select max(e2.commission_pct) from employees e2 where e2.manager_id = e.manager_id);
No 9 SS (of Query & Output in Shell)	
No 10 Query (as Plain Text)	select job_id, avg(salary) as Avg_Salary from employees group by job_id Having avg(salary) < (select avg(salary) from employees);

No 10 SS (of Query & Output in Shell)	

```
MariaDB [Company_24101204]> select job_id, avg(salary) as Avg_Salary
    -> from employees
    -> group by job_id
    -> Having avg(salary) < (select avg(salary) from employees);
+--------+------------+
| job_id | Avg_Salary |
+--------+------------+
| FI_ACC | 51500.0000 |
| HR_REP | 49000.0000 |
| SA_REP | 57666.6667 |
+--------+------------+
3 rows in set (0.001 sec)
```